

Cell Birth, Cell Death & Tissue Homeostasis

Current 68-page faculty deck primary - official Week 4 objectives integrated - Rb restriction-point control, DNA-content flow cytometry, p53 decisions, and intrinsic/extrinsic apoptosis

6 official objectives

Rb/E2F

Flow cytometry

p53/p21/Bax

Bcl-2 family

Caspases

SOURCE PRIORITY: the uploaded current faculty deck controls terminology, examples, and emphasis. The official Week 4 learning-objective sheet defines what must be mastered. This companion reorganizes the same material for retrieval and application without replacing the source deck.

Official Week 4 Objectives

1. Explain how retinoblastoma (Rb) protein controls the mammalian cell cycle and how loss of Rb function contributes to human disease.
2. Describe the molecular basis of flow cytometry and use DNA content to identify cell-cycle status.
3. Describe p53 regulation of the G1 checkpoint and distinguish p53-driven growth arrest from apoptosis.
4. Explain the intrinsic pathway of apoptosis and the roles of mitochondria, caspases, and Bcl-2 family proteins including Bax, Bak, Bid, and Bim.
5. Identify morphological, biochemical, and molecular features of apoptosis and techniques used to detect it.

6. Describe the extrinsic pathway of apoptosis and contrast it with the intrinsic pathway.

Slides 3-7

Tissue homeostasis is a balance between proliferation and cell death

Overview

Tissues maintain stable structure and function by balancing production of new cells with elimination of damaged, unnecessary, or dangerous cells. Growth-factor signaling pushes cells toward proliferation; checkpoint and death pathways prevent damaged cells from propagating.

Lecture organizing idea: Rb connects extracellular growth signals to the restriction point, while p53 responds to internal damage and can choose cytostasis or cytotoxicity.

Slides 8-13

Rb restrains E2F until growth signals activate Cyclin D-CDK4/6

LO 1

growth factor signaling

->

Cyclin D + CDK4/6

->

Rb phosphorylation/inactivation

->

E2F released

->

S-phase genes

Active, hypophosphorylated **Rb binds E2F** and blocks G1-to-S progression. Growth-factor-dependent CDK activity phosphorylates Rb, releases E2F, and lets the cell pass the restriction point.

Tumor-suppressor logic: loss of both functional Rb alleles removes this brake. Tumors can also functionally inactivate Rb by overactivating Cyclin D-CDK4/6. This is why CDK4/6 inhibitors can restore a cell-cycle brake in selected cancers.

Slides 14-19

Flow cytometry estimates cell-cycle phase from DNA content

LO 2

Peak/region	DNA content	Interpretation
G1	2N	Pre-replication
S	Between 2N and 4N	DNA synthesis
G2/M	4N	Replication complete; before cytokinesis
Sub-G1	<2N	Fragmented DNA; can mark apoptotic population

Fixed cells are stained with **propidium iodide**, which intercalates into DNA. Fluorescence intensity is proportional to cellular DNA content. The lecture uses CDK4/6 inhibition to illustrate G1 accumulation and taxol to illustrate G2/M arrest followed by mitotic death.

Slides 20-28

p53 converts DNA damage into either growth arrest or apoptosis

LO 3

Lower damage signal

p53 induces **p21**, a CDK inhibitor → G1 arrest and time for repair. This is a **cytostatic** response.

Higher/persistent damage

p53 induces pro-apoptotic effectors such as **Bax** → mitochondrial permeabilization and cell death. This is a **cytotoxic** response.

p53 is a tumor suppressor: its activation blocks propagation of damaged genomes. Loss of p53 removes both the repair pause and an important route to apoptosis.

Slides 28-32, 42-46

Intrinsic apoptosis is a mitochondrial caspase cascade controlled by Bcl-2-family balance

LO 4

cell stress / p53

->

Bax/Bak activation

->

mitochondrial outer-membrane permeabilization

->

cytochrome c release

->

caspase-9

->

caspase-3/6/7

Family role	Examples	Effect
Anti-apoptotic	Bcl-2, Bcl-XL, Mcl-1	Bind/inhibit pro-death members; preserve mitochondrial membrane
Pro-apoptotic	Bax, Bak, Bid, Bim	Promote mitochondrial permeabilization or activate pro-death machinery

Cancer connection: Bcl-2 overexpression raises the threshold for intrinsic apoptosis and can let damaged cells survive.

Slides 33-47

Apoptosis has recognizable membrane, nuclear, and DNA signatures

LO 5

Membrane

Plasma membrane remains intact until late, but **phosphatidylserine (PS)** flips to the outer leaflet, flagging the cell for engulfment.

Nucleus

Chromatin condenses and fragments; nuclear lamins and other structural proteins are cleaved by executioner caspases.

DNA

Caspase-activated endonucleases cut chromatin into nucleosomal fragments, producing a ladder-like pattern and potentially a sub-G1 flow peak.

Assay	What it detects
Annexin V	Externalized phosphatidylserine
Propidium iodide	Loss of membrane integrity / DNA content depending assay context
Cleaved-caspase / cleaved-lamin antibodies	Proteolytic execution of apoptosis

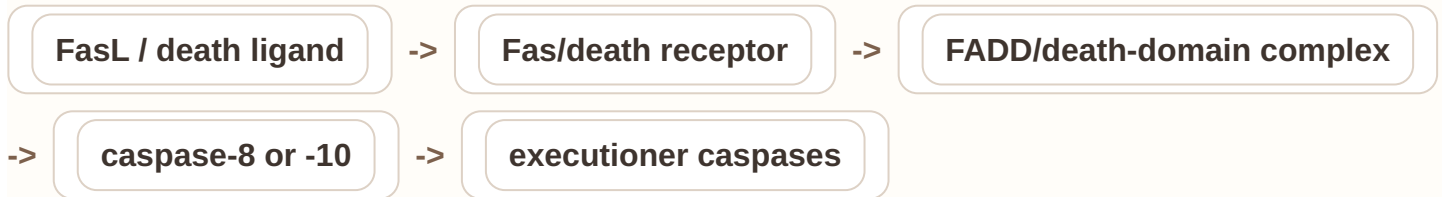
DNA fragmentation / laddering

Endonuclease-mediated nucleosomal cleavage

Slides 49-52, 68

Extrinsic apoptosis begins at a death receptor rather than the mitochondrion

LO 6



Cross-talk: caspase-8 can cleave Bid to **tBid**, which activates the mitochondrial intrinsic pathway. Thus extrinsic signaling can amplify itself through mitochondria.

Intrinsic	Extrinsic
Internal stress / DNA damage	External death-ligand signal
Mitochondrial cytochrome c central	Death receptor/FADD central
Initiator caspase-9	Initiator caspase-8/10

Slides 38-48

Apoptosis and necrosis differ in membrane integrity and inflammatory consequence

LO
5

Apoptosis	Necrosis
Cell shrinkage, chromatin condensation, apoptotic bodies	Cell/organelle swelling and membrane rupture
Membrane preserved until late; PS externalized	Membrane compromised early
Ordered DNA fragmentation / laddering	More random DNA degradation; no classic ladder
Efficient phagocytic clearance; less inflammatory	Cell contents spill into tissue; inflammatory response

Exam traps

Protect checkpoint, cytometry, and caspase relationships

All LOs

Trap 1: Active Rb binds E2F and blocks S phase; phosphorylated/inactive Rb releases E2F.

Trap 2: G1 = 2N; G2/M = 4N; S lies between.

Trap 3: p21 mediates p53 growth arrest; Bax helps mediate p53 apoptosis.

Trap 4: Intrinsic initiator = caspase-9; extrinsic initiators = caspase-8/10.

Trap 5: Bcl-2 is anti-apoptotic; Bax/Bak are pro-apoptotic.

Trap 6: Annexin V recognizes externalized phosphatidylserine.

30-SECOND FINISH

Growth factor → Cyclin D/CDK4/6 → Rb OFF → E2F ON. PI flow: G1 2N, S 2-4N, G2/M 4N. p53 low damage → p21 arrest; high damage → Bax death. Intrinsic: mitochondria/cytochrome c/caspase-9. Extrinsic: Fas/FADD/caspase-8/10. Both converge on executioner caspases.